

Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: GLUE | Context: Luxembourgish Public Administration Civil Servants — GLUE Assessment

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GLUE evaluates text-based outputs only via hard single-label scoring on private test sets [Q57, Q58], with per-task scores aggregated as a macro-average [Q59, Q60] subject to two-submissions-per-day rate limiting [Q131]. The deployment is also text-in/label-out, so the surface-level format aligns — both are written-text pipelines, and OF is correctly marked MODERATE priority (lower than HIGH for the other dimensions). However, GLUE's output form lacks the multi-label structure and confidence scoring the deployment requires (elicitation A4), and the submission-based privately-held test data model [Q7, Q58] prevents the iterative, task-specific calibration a live civil-service deployment requires. The macro-average [Q60] also rewards balanced performance across nine irrelevant English NLU tasks, making the aggregate GLUE score uninformative as a deployment proxy. Score is 2 (not 1) because the underlying text-based output modality matches and the per-task evaluation infrastructure is methodologically reusable, but the specific output-form requirements (multi-label, confidence) are unsatisfied.

ID	Evidence
Q57	"The GLUE benchmark follows the same evaluation model as SemEval and Kaggle." (p.5)
Q58	"To evaluate a system on the benchmark, one must run the system on the provided test data for the tasks, then upload the results to the website gluebenchmark.com for scoring." (p.5)
Q59	"The benchmark site shows per-task scores and a macro-average of those scores to determine a system's position on the leaderboard." (p.5)
Q60	"For tasks with multiple metrics (e.g., accuracy and F1), we use an unweighted average of the metrics as the score for the task when computing the overall macro-average." (p.5)
Q131	"The GLUE website limits users to two submissions per day in order to avoid overfitting to the private test data." (p.14)
Q7	"Four of the datasets feature privately-held test data, which will be used to ensure that the benchmark is used fairly." (p.1)

Strengths

- Output modality matches: GLUE produces text-based predictions [Q57, Q58], aligned with the deployment's text-in/text-or-label-out pipeline.
- Per-task scoring with task-appropriate metrics (accuracy, F1, MCC, Pearson [Q20, Q23, Q30, Q35]) is methodologically transferable to a custom benchmark suite, even though the specific tasks are not.
- Submission rate-limiting [Q131] models a defensive-evaluation practice (preventing test-set overfitting) potentially adaptable to civil-service contexts.

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Q58	"To evaluate a system on the benchmark, one must run the system on the provided test data for the tasks, then upload the results to the website gluebenchmark.com for scoring." (p.5)
Q20	"Unless otherwise mentioned, tasks are evaluated on accuracy and are balanced across classes." (p.3)
Q23	"Following the authors, we use Matthews correlation coefficient (Matthews, 1975) as the evaluation metric, which evaluates performance on unbalanced binary classification and ranges from -1 to 1, with 0 being the performance of uninformed guessing." (p.3)
Q30	"Because the classes are imbalanced (68% positive), we follow common practice and report both accuracy and F1 score." (p.3)

Q35	"The Semantic Textual Similarity Benchmark (Cer et al., 2017) is a collection of sentence pairs drawn from news headlines, video and image captions, and natural language inference data. Each pair is human-annotated with a similarity score from 1 to 5; the task is to predict these scores. Follow common practice, we evaluate using Pearson and Spearman correlation coefficients." (p.3)
Q131	"The GLUE website limits users to two submissions per day in order to avoid overfitting to the private test data." (p.14)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Partial match. Output modality (text labels) matches the deployment's text-classification pipeline. However, the specific output form — single hard labels, no multi-label, no confidence scores — does not match the deployment's multi-label-plus-confidence requirement (elicitation A4, DATASET-Concern 4).

OF-2: Not applicable — neither GLUE nor the deployment require speech-based output. The deployment is text-classification-only per the elicitation.

OF-3: Not directly relevant to OF for this deployment (the deployment's end-users are civil servants, not citizens; outputs are routing decisions and flags consumed by an internal review interface). Literacy/accessibility considerations would apply to citizen-facing response generation, which is outside the elicited scope.

OF-4: Form mismatches that harm external validity: (a) GLUE's macro-average over nine English tasks [Q59, Q60] is not informative as a deployment proxy; (b) hard single-label scoring [Q12] cannot evaluate multi-label routing; (c) the privately-held test-set submission model [Q7, Q58] prevents task-specific calibration; (d) two-submissions-per-day limit [Q131] is misaligned with iterative civil-service development cycles.

ID	Evidence
Q12	"All tasks are single sentence or sentence pair classification, except STS-B, which is a regression task. MNLI has three classes; all other classification tasks have two. Test sets shown in bold use labels that have never been made public in any form." (p.2)
Q57	"The GLUE benchmark follows the same evaluation model as SemEval and Kaggle." (p.5)
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Information Gaps

- Whether the deployment requires structured output beyond labels + confidence (e.g., extracted entities for NER, span annotations) is mentioned in the deployment description but not detailed in OF terms.

Requires Expert Verification

- Specific output-schema requirements for the routing API (label space, confidence-threshold rules, NER span format) should be confirmed with deployment engineers.

Remediation

Gap 1: GLUE's macro-average over nine English tasks [Q60] is uninformative as a deployment proxy, and the privately-held submission model [Q7, Q58] prevents iterative calibration.

Recommendation: Avoid using the GLUE macro-average as any kind of deployment indicator. Build an internal evaluation infrastructure mirroring GLUE's per-task scoring [Q59] but with task-specific thresholds aligned to civil-service operational needs, fewer submission-rate constraints, and integration with EU AI Act high-risk-system documentation requirements [WEB-22].

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WEB-22	EU AI Act Annex III — automated administrative routing systems classified high-risk; requires output-form transparency including confidence-scored decisions (https://artificialintelligenceact.eu/annex/3/)